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Naringenin inhibits the microsomal triglyceridetransfer protein/apolipoprotein B axis to inhibit intestinal metaplasia progression

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Abstract

Intestinal metaplasia (IM) is a premalignant condition that increases the risk for subsequent gastric cancer (GC). Traditional Chinese medicine generally plays a role in the treatment of IM, and the phytochemical naringenin used in Chinese herbal medicine has shown therapeutic potential for the treatment of gastric diseases. However, naringenin's specific effect on IM is not yet clearly understood. Therefore, this study identified potential gene targets for the treatment of IM through bioinformatics analysis and experiment validation. Two genes (*MTTP* and *APOB*) were selected as potential targets after a comparison of RNA-seq results of clinical samples, the GEO dataset (GSE78523), and naringenin-related genes from the GeneCards database. The results of both cell and animal experiments suggested that naringenin can improve the changes in the intestinal epithelial metaplasia model via *MTTP/APOB* expression. In summary, naringenin likely inhibits the *MTTP/APOB* axis and therefore inhibits IM progression. These results support the development of naringenin as an anti-IM agent and may contribute to the discovery of novel IM therapeutic targets.

KEYWORDS

bio-informatics analysis, gastric cancer, intestinal metaplasia, naringenin, traditional Chinese medicine

Xiangming Huang, Mengqiu Zhang, Lina Gu, and Ziyang Zhou contributed equally to this study.

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1 | INTRODUCTION

Intestinal metaplasia (IM) is characterized by the pathological transformation of non-intestinal epithelium into an intestinal mucosa (Chiu et al., 2014; Rubio & Befrits, 2004). Such lesions are commonly regarded as precursors of gastric cancer (GC), a leading global cause of cancer-related deaths (Nagtegaal et al., 2020; Zhang et al., 2021). The pathophysiological process of the transition of mucosa becoming cancerous through IM has been established as the classic Correa Cascade reaction (Correa, 1992). Patients with IM have an elevated risk of progression to GC, with an annual incidence rate of 0.25% and meta-analyses have indicated that IM increases the risk of GC by 3.6-fold (de Vries et al., 2008; Shao et al., 2018), it is important to identify high-risk patients with IM to reduce the likelihood of advancement to GC after IM.

In the management of IM in modern medicine, bactericidal therapy is considered the primary treatment option when IM coexists with *Helicobacter pylori* infection (Altayar et al., 2020; Kang et al., 2023; Kumar et al., 2020; Reyes, 2023; Sue et al., 2015; Thrift & El-Serag, 2020). Additionally, non-specific therapeutic approaches such as gastric mucosal protective agents, gastrointestinal motility modulators, and vitamin supplementation are frequently employed (Abosamra et al., 2023; Chen et al., 2010; Sipponen & Maaroos, 2015). In recent years, traditional Chinese medicine such as Mo-Luo-Dan has become increasingly popular in the treatment of IM (Lou et al., 2024; Tang et al., 2016; Zhou et al., 2022). The Qi-Ling-Fang (QLF) has also demonstrated favorable therapeutic outcomes (Lumin & Wangdeming, 2010; Zhang et al., 2023). Comprising six traditional Chinese medicines, named *Astragalus membranaceus* (Fisch.) Bunge (HQ), *Ganoderma lucidum* (Leyss.exFr.), Karst (LZ), Semen Coicis (YYR), *Citrus reticulata* Blanco. (CP), Herba *Hedyotis diffusa* (BHSSC), and Herba *Salviae chinensis* (SJC), this prescription exhibits commendable clinical efficacy despite an unclear treatment mechanism. However, literatures exploring possible mechanisms on QLF's effectiveness for treating IM are still lacking.

Therefore, in this study, we analyzed different gene expressions in patients with IM using clinical samples and data from a public database. Additionally, we also identified a gene set correlated with the progression of IM to GC. When comparing the target genes of QLF with the hub genes identified as pivotal for the development of IM, we found two genes that may play important roles in the treatment of IM and validated these mechanisms using animal and human cell experiments.

2 | MATERIALS AND METHODS

2.1 | Identifying the active ingredients of QLF and target genes prediction

The active ingredients of QLF were assessed using the Traditional Chinese Medicine Systematic Pharmacology Database and Analysis Platform (TCMSP, <http://tcmssp.com/tcmssp.php> [Ru et al., 2014]).

To determine the eligible active ingredients, we used oral bioavailability (OB) of 30% and drug-likeness (DL) of 0.18 as screening criteria. By using the Uniprot platform (<https://www.uniprot.org> [Wang et al., 2021]), the target proteins of these active ingredients were downloaded and converted into gene symbols.

2.2 | Sample preparation and RNA sequencing

This study was approved by the institutional review board of Nanjing Medical University (IRB number: 2022-254) and was conducted in accordance with the Declaration of Helsinki. Clinical samples and clinical records were used with informed consent.

Gastric biopsies were collected from chronic atrophic gastritis (CAGs) patients with or without IM diagnoses at the Nanjing Medical University Affiliated Cancer Hospital prior to the administration of any medications. Total RNA was then extracted using TRIzol (Invitrogen, Thermo Fisher Scientific, Waltham, MA, USA) and stored immediately at -80°C for further analysis. RNA-sequencing was performed by using IlluminaNovaseq™6000 (LC-Bio Technologies (Hangzhou) Co., Ltd., China).

2.3 | Data collection from Gene Expression Omnibus (GEO) database

The GEO database was also used to obtain gene expression profiles related to IM (Barrett et al., 2013). Based on the GPL18990 platform (Affymetrix Human Almac Xcel Array), GSE78523 contains gastric mucosa samples from 15 healthy donors and 30 IM patients (Companiononi et al., 2017). The Series Matrix File of GSE78523 was obtained from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE78523>. The annotation file of the GPL18990 platform was obtained from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GPL18990>.

2.4 | Differentially expressed genes (DEGs) identification and functional enrichment analysis

DEGs from IM patients and healthy controls were screened using "DESeq2" in R. DEGs met the criteria of $|\log \text{fold change (FC)}| > 2$ and an adjusted p -value < 0.05 was considered significant.

Gene ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were conducted using the "clusterProfiler (v3.14.3)" package (Ashburner et al., 2000; Kanehisa & Goto, 2000). A cut-off of p -adj < 0.05 and q value < 0.2 was used for KEGG and GO terms, respectively.

To identify which signaling pathways are enriched in IM patients, gene set enrichment analysis (GSEA) was performed (Subramanian et al., 2005). A subset of genes (c2.cp.kegg.v7.5.1.symbols.gmt) from the Molecular Signatures Database (MSigDB) (<http://www.gsea-msigdb.org/gsea/downloads.jsp>) was used as a reference set (Liberzon et al., 2011), and other parameters were set according to

defaults. Statistical significance was established when $p < 0.05$, the normalized enrichment score (NES) exceeded 1, and the false discovery rate (FDR) was below 0.25.

2.5 | Weighted gene co-expression network analysis (WGCNA)

Gene clusters correlated with clinical characteristics were identified through WGCNA, an R package (Langfelder & Horvath, 2008). Scale-free networks are used to calculate a suitable soft threshold. We then computed the dissimilarity (1-TOM) of the weighted adjacency matrix, which was derived from a topological overlap matrix (TOM). Dendrograms were colored according to the modules assigned by dynamic cuts. A selection of modules most relevant to the clinical phenotype of the risk score was then selected for further analysis.

2.6 | Protein-protein interaction (PPI) network construction and analysis

According to the Search Tool for Retrieval of Interacting Genes (STRING) database (<http://string-db.org> [Szklarczyk et al., 2023]), a PPI network was constructed with the combined score = 0.4 used as a threshold (Szklarczyk et al., 2023). We further analyzed the network using Cytoscape software (version 3.8.2, Cytoscape Consortium, San Diego, CA, USA) (Majeed & Mukhtar, 2023; Shannon et al., 2003). We then ranked each node according to several topological metrics, including maximal clique centrality (MCC), density of maximum neighborhood component (DMNC), maximum neighborhood component (MNC), and degree using the cytoHubba plug-in (Chin et al., 2014).

2.7 | Molecular docking analysis

The protonation state of the small molecule was set to pH = 7.4, and the compound was extended into a 3D structure using Open Babel. After obtaining the stereo structures of the proteins and compounds by Uniprot (Wang et al., 2021) and PubChem (Kim et al., 2021), respectively, the protein and compound structures were processed using AutoDock tool (ADT3). Subsequently, the protein-docking box parameters were calculated and generated through the AutoGrid program, and then molecular docking is performed using Autodock Vina (1.2.0) (Eberhardt et al., 2021). After the docking was completed, the results were visualized and analyzed using PyMol and Discovery Studio 2021 (Jiang et al., 2023; Seeliger & de Groot, 2010), respectively.

2.8 | Cell culture and treatment and transient transfection

GES-1 cells were purchased from Procell Life Science & Technology Co., Ltd, and cultured in RPMI1640 containing 10% fetal bovine

TABLE 1 Primer pair sequences.

Gene		Sequence (5'–3')
APOB	Forward	TGCTCCACTCACTTTACCGTC
	Reverse	TAGCGTCCAGTGTGTACTGAC
CDX2	Forward	GACGTGAGCATGTACCCTAGC
	Reverse	GCGTAGCCATTCCAGTCCT
MTTP	Forward	ACAAGCTCACGTACTCCACTG
	Reverse	TCCTCCATAGTAAGGCCACATC
MUC2	Forward	GAGGGCAGAACCCGAAACC
	Reverse	GGCGAAGTTGTAGTCGCAGAG
GAPDH	Forward	AACAGCCTCAAGATCATCAGCA
	Reverse	ATGAGTCCTTCCACGATACCA

serum (FBS) and 1% penicillin at 37°C with 10% CO₂. All the cells tested were from 3rd to 8th generations. GES-1 cells were seeded in a 6-well plate at 2×10^5 cells per well and transfected with lentiviral vectors knocking down microsomal triglyceridtransfer protein (MTTP). The lentiviral vectors were obtained from GenePharma Company. The target sequence was as follows: GCGTATTAGCAG-GATGTGAAT. Deoxycholic acid (DCA) (HY-N0593A) and Naringenin (HY-W011641, purity 98.88%) were purchased from MedChemExpress (Monmouth Junction, NJ, USA).

2.9 | RNA extraction and quantitative real-time PCR (qRT-PCR)

In accordance with manufacturer instructions (Xuan et al., 2023), total RNA was extracted from GES-1 cells using SteadyPure Quick RNA Extraction Kit (AG21023, Accurate Biotechnology, Hunan Co., Ltd, China). As a reaction preparation, we used HiScript[®] III RT SuperMix for qPCR (R323, Vazyme, Nanjing, China) for reverse transcription. The qRT-PCR was then performed using ChamQ Universal SYBR qPCR Master Mix (Q711, Vazyme, Nanjing, China). The following conditions were followed: Stage 1: 95°C, 30 s, 1 cycle; Stage 2: 95°C, 5 s → 60°C, 30 s, 40 cycles. An ABI 7500 real-time PCR instrument (Applied Biosystems, Foster City, CA, USA) was applied for the reaction. GAPDH served as a normalization factor in the computational calculation of gene expression using the comparative threshold (CT) method ($\Delta\Delta CT$). In Table 1, we list the primer pairs used for qRT-PCR.

2.10 | Animal experiment

We used 4- to 6-week-old male Sprague-Dawley (SD) rats weighing 150 ± 30 g for experiments (Watanabe & Ito, 1986; Xie et al., 2024; Zhu et al., 2024). The rats were raised under standard laboratory conditions include a 12-h light/dark cycle, temperature between 20 and 26°C, and humidity between 40% and 70%. The rats were free to access food and water. After an

adaptation period of 7 days, the 60 rats were randomly divided into four groups: the control group ($n = 15$), model group ($n = 15$), naringenin-treated group ($n = 15$), and moluodan-treated group ($n = 15$) as positive control (Lou et al., 2024; Tang et al., 2016; Zhou et al., 2022). Except for the control group, the rats in the other groups were orally gavaged with 0.02 mol/L MNNG solution at a dose of 0.5 mL/100 g/day using an auxiliary method of disrupted feeding to induce the model. Since the 20th week, one rat every week from each group was sacrificed in order to validate if the model was successfully established. After confirming that IM model was successful, the remaining rats in the naringenin-treated group and moluodan-treated group was additionally gavaged with 50 mg/kg/day naringenin or 10 mL/kg/day moluodan for 8 weeks. The other rats received normal saline dosages every day intragastrically. The animal experiment was approved by the institutional review board of School of Third Clinical Medicine, Nanjing University of Chinese Medicine. (AEWC number: 20220715-223).

2.11 | Preparation of gastric tissue and serum

Rats were anesthetized by intraperitoneal injection of 0.2 mL/100 g pentobarbital (3%). Serum samples were collected from the abdominal aorta for enzyme-linked immunosorbent assays (ELISA). The stomach was then removed for immunohistochemistry analyses.

2.12 | Immunohistochemistry

The stomach tissues were fixed in 4% paraformaldehyde overnight, and embedded in paraffin. Paraffin sections were heated in a pressure cooker with 10 mM citrate buffer (pH 6.0) for 4 min for antigen retrieval. The sections were then treated in 3% H₂O₂ at room temperature for 30 min to eliminate endogenous peroxidase. The sections were then incubated overnight at 4°C with rabbit anti-caudal-type homeobox 2 (CDX2) antibody (1:100, Aifang, Hunan, China), rabbit anti-mucin 2 (MUC2) antibody (1:100, Aifang, Hunan, China), rabbit anti-apolipoprotein B (APOB) antibody (1:100, Proteintech, Wuhan, China), or rabbit anti-microsomal triglyceride transfer protein (MTTP) antibody (1:100, Yi Fei Xue, Nanjing, China). Subsequently, they were incubated with biotinylated goat anti-rabbit IgG (1:200, Aifang, Hunan, China) in PBS, and then incubated in avidin-biotin-peroxidase complex (1:100, Aifang, Hunan, China) in phosphate buffered saline (PBS) at room temperature for 1 h, followed by incubation in 0.05% 3,3'-diaminobenzidine and 0.01% H₂O₂ in 0.1 M PBS for 5 min for visualization. The immuno labeled sections were dehydrated in a graded ethanol series, defatted in xylene, and mounted. Control experiments followed a similar procedure, except the sections were processed either in the presence of antibodies pre-adsorbed overnight with excess immunizing recombinant protein or in the absence of the primary CDX2 antibody. At least three field images were analyzed per sample.

TABLE 2 Clinic characteristics of intestinal metaplasia (IM) patients included in this study.

ID	Gender	Age
CTRL1	Female	35
CTRL2	Male	43
CTRL3	Male	43
IM1	Female	71
IM2	Male	80
IM3	Female	62

2.13 | ELISA analysis

The levels of CDX2, MUC2, MTTP, and APOB in serum and gastric tissue samples were detected according to the manufacturer's protocols (Aifang, Hunan, China). All samples were run in duplicates. Data were collected using a Rayto RT-6100 analyzer at a wavelength of 450 nm.

2.14 | Statistical analysis

Data analysis for RNA-seq was carried out using R Studio (version 1.4.1717, R studio, Boston, MA, USA) software with R version 4.1.2. GraphPad PRISM (version 9.0) was used to analyze and visualize the data for in vitro experiments. *T*-tests (equal variance) and corrected *t*-tests (unpaired test corrected for Welch's correction, unequal variance) were used to compare the two groups. Statistical significance was determined by a *p*-value of 0.05 for all tests, and a minimum of three independent replicates were conducted for all in vitro experiments.

3 | RESULTS

3.1 | Screening of active compounds and potential targets of QLF

The active compounds and potential targets of the drugs were downloaded from TCMSP database. A total of 43 active ingredients were selected, which related to 248 target genes. Construction of herb-gene-disease network was subsequently done to assess the work model of these drugs (Supplemental Figure 1).

3.2 | RNA-sequencing profiles revealed a distinct gene signature in patients with IM

Between September and December of 2022, six patients with CAG were enrolled in our study. Among them, three were diagnosed with CAG with IM, while the other three had CAG only. Clinical

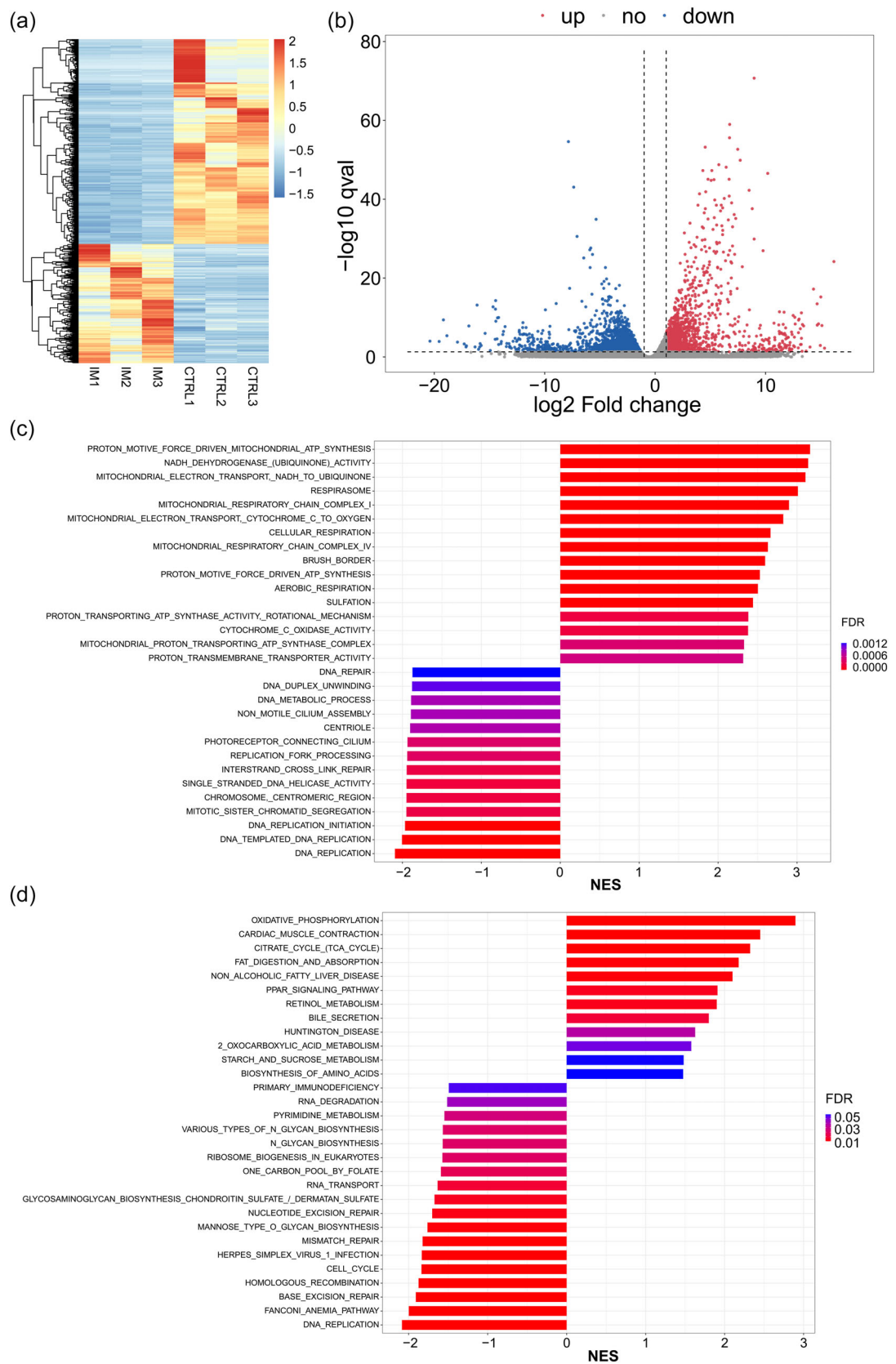


FIGURE 1 (a) Heatmap of gene expression of CAG patients. (b) 2685 eligible genes. (c, d) GSEA analysis.

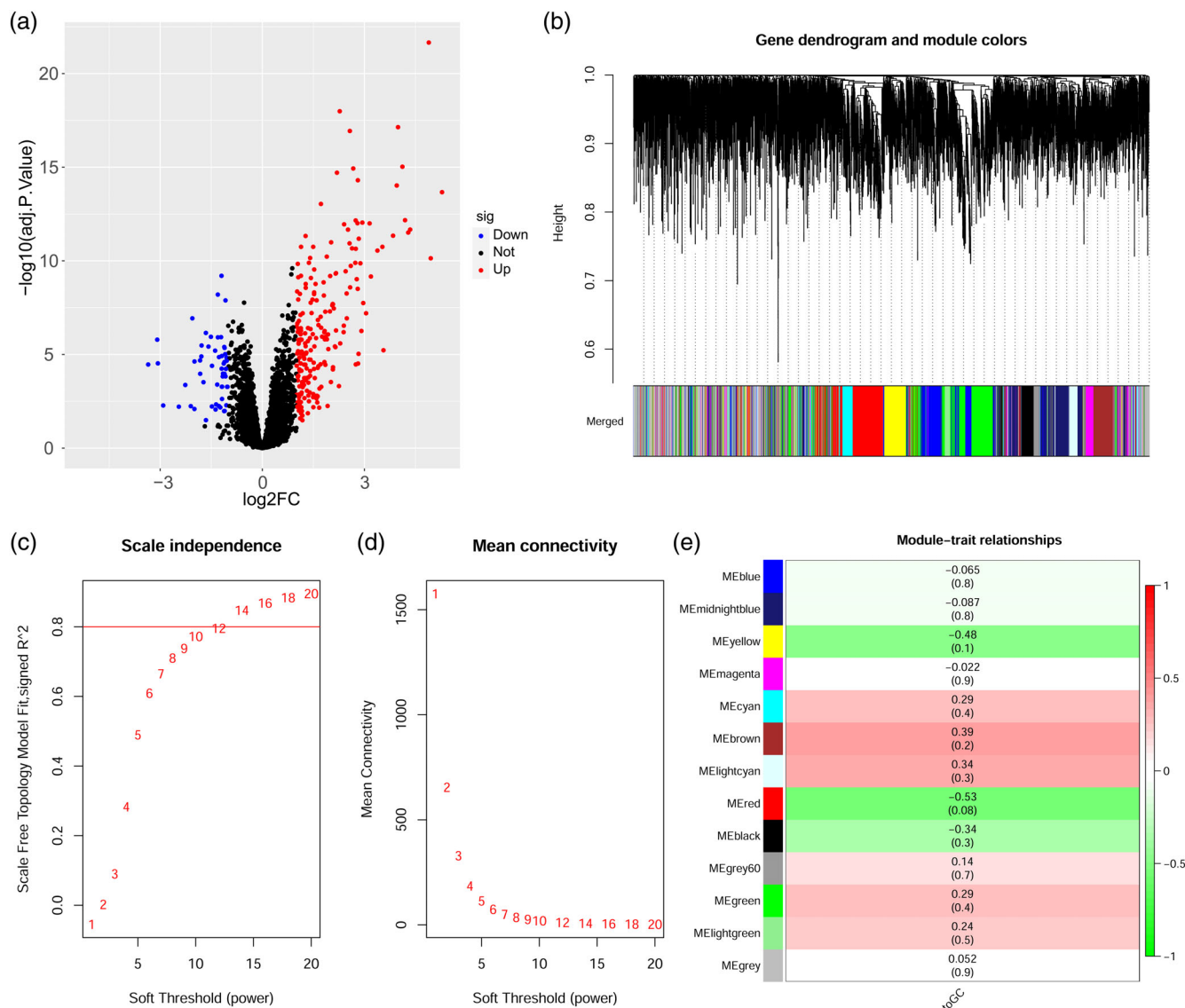


FIGURE 2 WGCNA indicated the co-expression modules.

characteristics of the participants involved in this study are summarized in Table 2.

A heatmap of gene expression was created for IM patients (Figure 1a). We first screened for DEGs. Using the criteria outlined above, we identified 2685 eligible genes, 1865 of which were down-regulated and 820 of which were up-regulated (Figure 1b).

We performed a GSEA analysis using the complete expression data set. The GSEA analysis included GO, KEGG, BIOCARTEA, PID, and REACTOME pathways. It indicated that CAG patients with IM showed that gene sets linked to bilateral breast cancer, breast disease, DNA replication, mitotic sister chromatid segregation, cell cycle, Fanconi anemia pathway, and association of RAD51 with RAD52. DNA double-strand break ends were under-expressed, while pathways linked to familial hypercholesterolemia, ATP synthesis, NADH dehydrogenase (ubiquinone) activity, mitochondrial respiratory chain

complex I, citrate cycle, oxidative phosphorylation, and electron transfer from reduced cytochrome C to molecular oxygen were over-expressed (Figure 1c, d, Supplementary Figure 2).

3.3 | DEGs combined with WGCNA methods for screening key IM genes

To identify genes involved in the development of IM, the GSE78523 dataset was analyzed. The sequencing data of 15 normal controls and 30 patients with IM were compared, and 69 differential genes (8 downregulated and 61 upregulated) were selected based on $|\log_2\text{FC}| > 2$ and adjusted p -value < 0.05 (Figure 2a).

While the most commonly identified gastric precancerous lesion is IM, factors controlling IM to GC progression remain

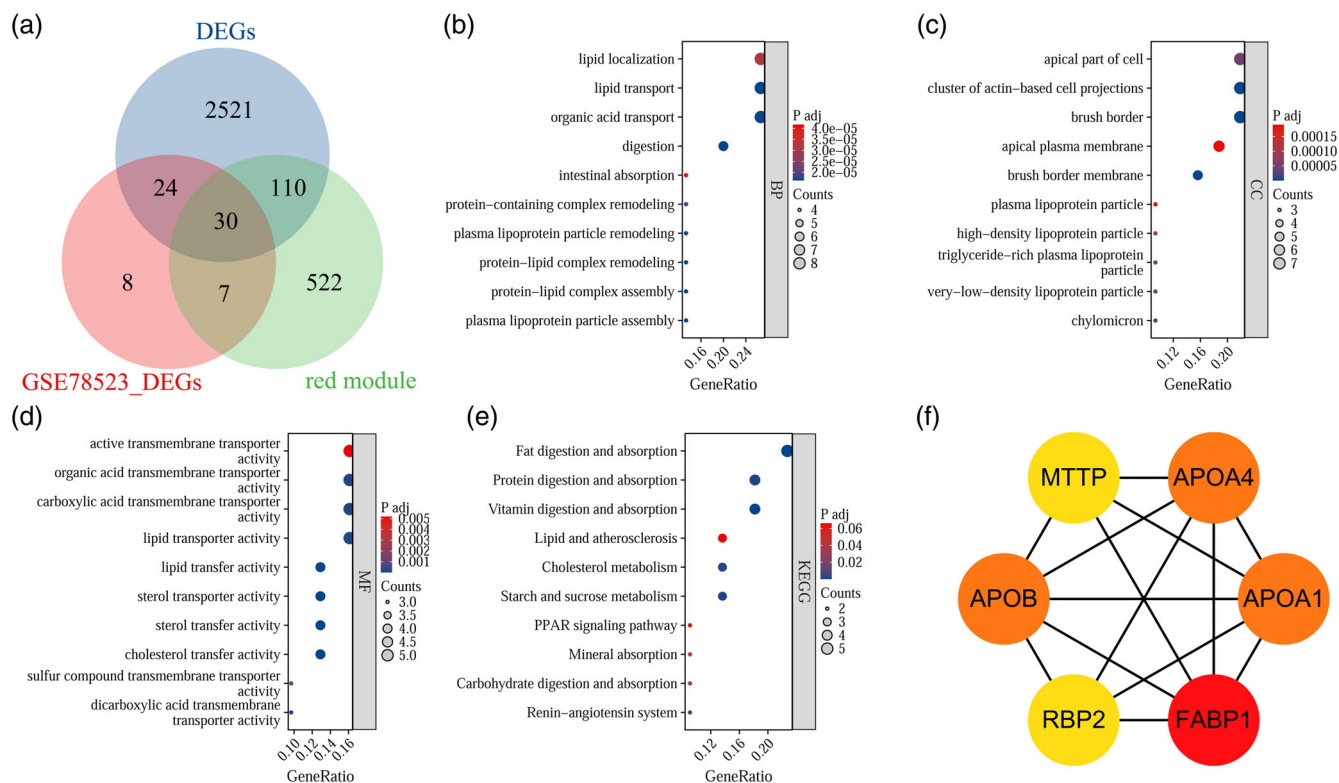
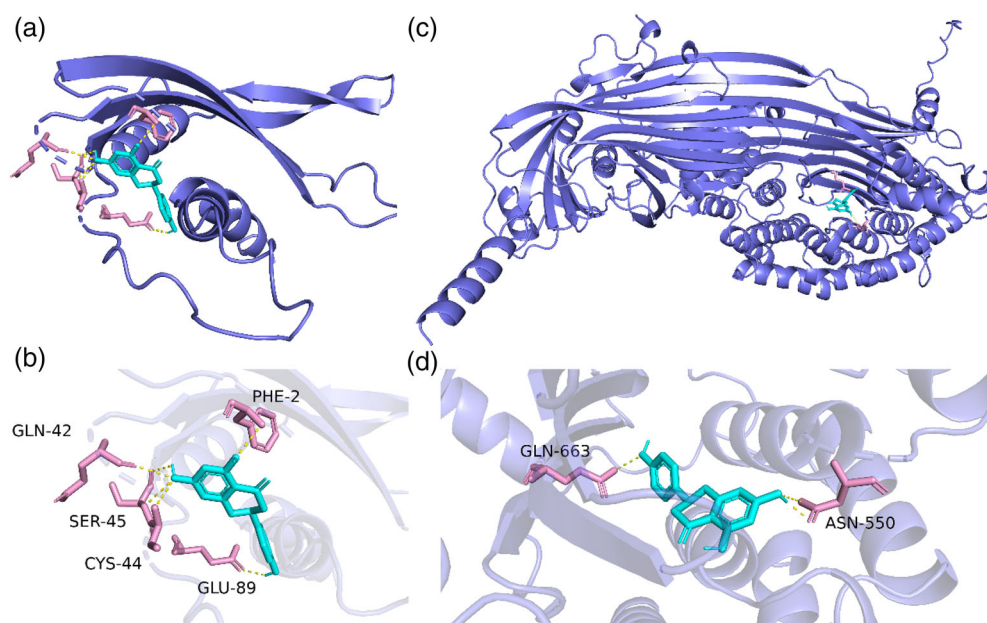


FIGURE 3 (a) Correlated genes with IM progression. (b–e) GO/KEGG pathway enrichment analysis. (f) Hub genes of IM.

FIGURE 4 Molecular docking analysis. (a, b) Simulate interactions of naringenin with APOB. (c, d) Simulate interactions of naringenin with MTP.



unknown. We used the WGCNA method to identify the gene set associated with this progression that exhibited the strongest correlation. The dataset GSE78523 contains RNA-seq results from 13 incomplete IM (IIM) patients of which 6 progressed to cancer after a follow-up period and 7 did not. In this study, the soft-thresholding power was chosen as 12 according to the scale-free

topology criterion (Figure 2b). Dynamic tree cut revealed 13 different gene modules in the hierarchical clustering dendrogram (Figure 2c). In Figure 2d, module-trait correlations show that multiple modules were associated with the progression from IM to GC, while MEred had a maximum negative correlation. Lastly, we selected genes from the red module for further analysis.

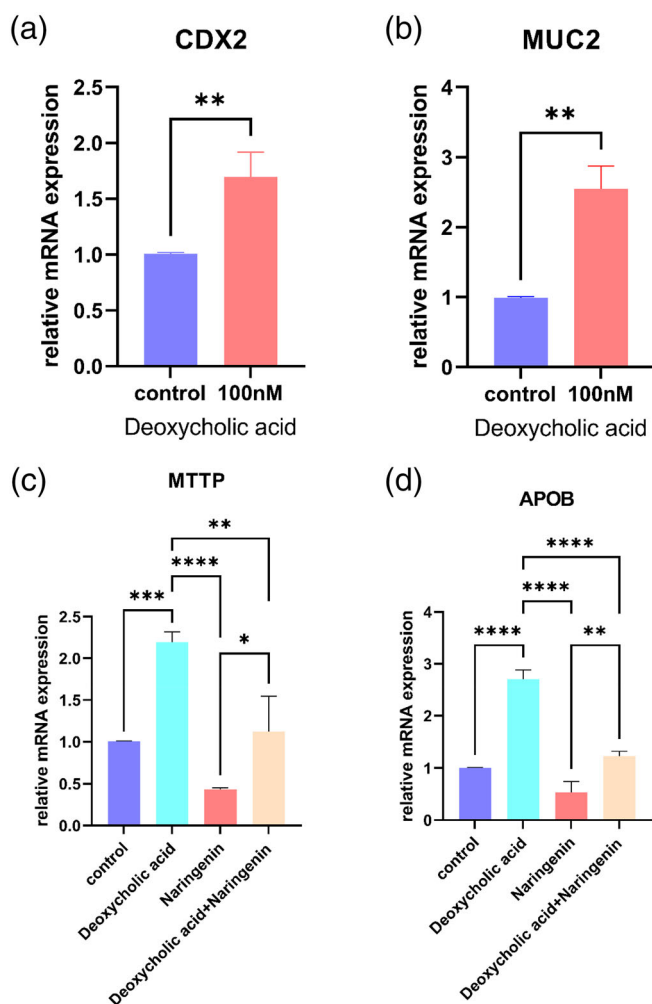


FIGURE 5 Naringin can ameliorate the IM model induced by DCA in GES-1 cells. (a, b) Total RNA was extracted from the DCA-induced IM model, and the expression of *CDX2* and *MUC2* was significantly increased, indicating a successful model induction (data were presented as the mean $\pm$ SEM. All experiments had been repeated at least three times). (c, d) Naringin can inhibit *MTTP*/*APOB* expression in the model group and improve the IM model induced by DCA in GES-1 cells (data were presented as the Mean $\pm$ SEM. All experiments had been repeated at least three times.). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: $P \geq 0.05$.

There were 54 common DEGs among clinical samples and the GSE78523 cohort. Among these, 30 genes were also contained in the red module which were correlated strongly with IM progression (Figure 3a). We assume that these 30 genes played important roles in the development of IM. GO/KEGG pathway enrichment analysis showed these genes to be mainly enriched in the following terms or pathways: organic acid transport, digestion, plasma lipoprotein particle assembly, protein-lipid complex assembly, cholesterol transfer activity, sterol transfer activity, lipid transfer activity, fat digestion and absorption, protein digestion and absorption, and cholesterol metabolism (Figure 3b–e). A PPI network of these 30 genes was constructed in the STRING database (Supplementary Figure 3) and analyzed with

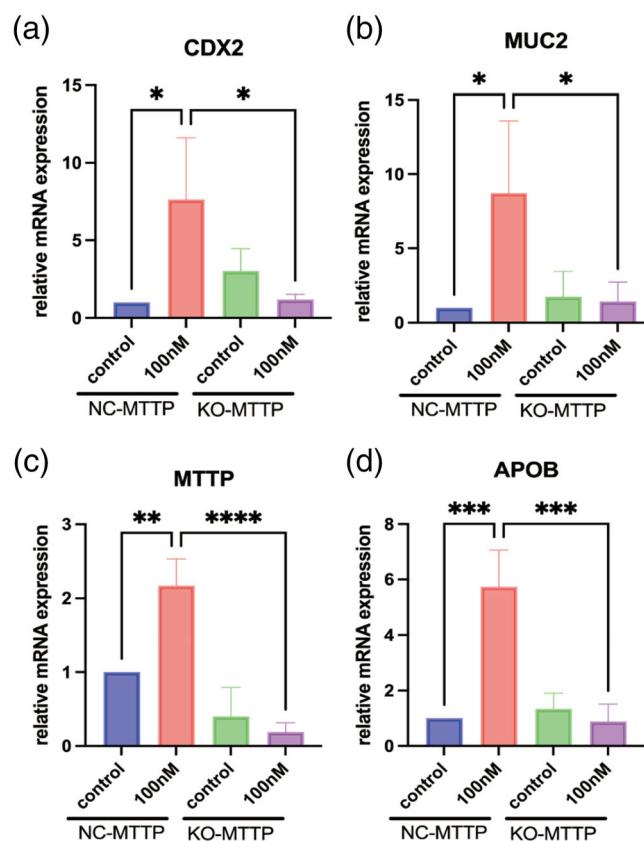


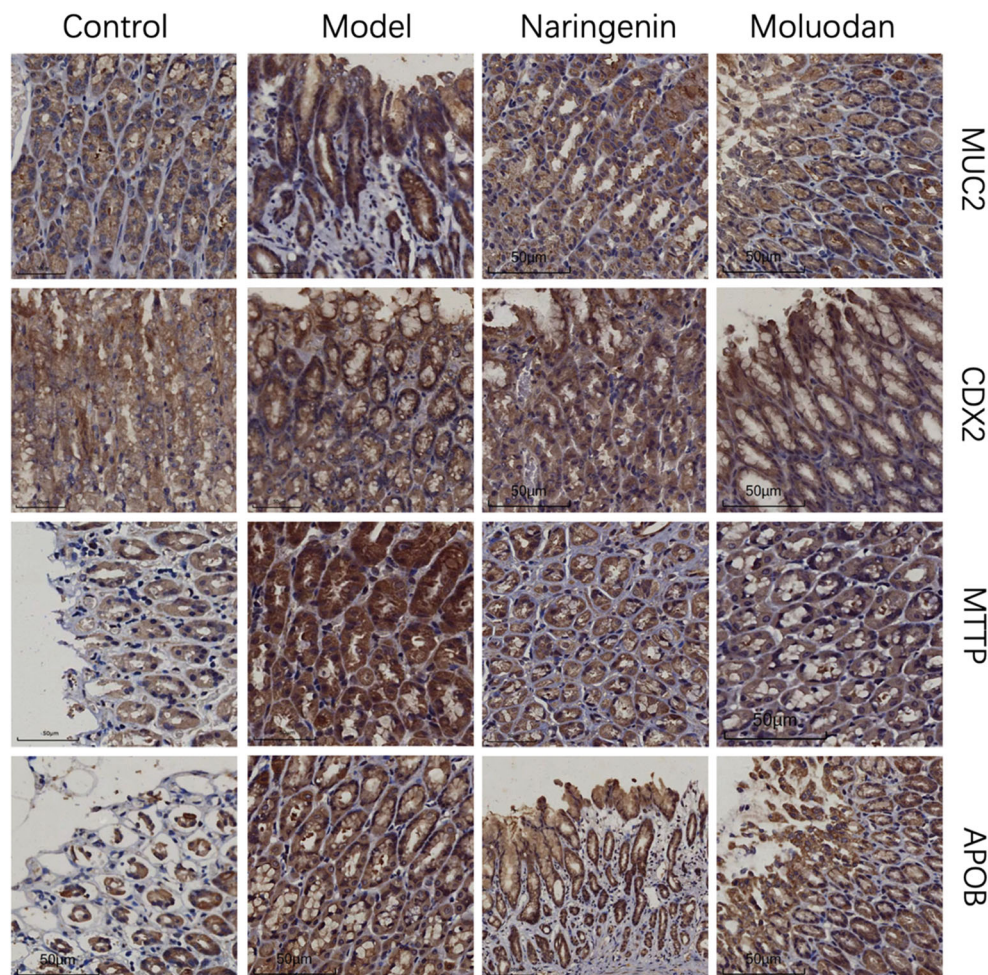
FIGURE 6 *MTTP* knockdown can improve the *MTTP*/*APOB* pathway induced by DCA in GES-1 cells. (a, b) Total RNA was extracted from the DCA-induced IM model, and the expression of *CDX2* and *MUC2* was significantly increased. After *MTTP* knockdown, the expression of *CDX2* and *MUC2* was significantly inhibited (data were presented as the mean $\pm$ SEM. All experiments had been repeated at least three times). (c, d) After *MTTP* knockdown, the expression of *MTTP* and *APOB* was significantly inhibited (data were presented as the Mean $\pm$ SEM. All experiments had been repeated at least three times). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: $P \geq 0.05$.

Cytoscape software. The MCC, DMNC, MNC, and Degree values for each gene were obtained and listed in Supplementary Table 1. In the table, *FABP1*, *APOA1*, *APOA4*, *APOB*, *MTTP*, and *RBP2* showed higher scores than the other genes and thus were identified as hub genes (Figure 3f). These results indicated that the six hub genes may play important roles in the development of IM.

3.4 | QLF regulated IM progression mainly through naringenin

To determine how QLF regulates IM, the target genes were obtained and a Venn plot was created comparing the targets of QLF and hub genes involved in IM progression. In Supplementary Figure 4, *APOB* and *MTTP* are located in the overlapped area. To our interest, both *APOB* and *MTTP* were targets of naringenin, an active compound

FIGURE 7 Immunoreactivity of CDX2, MUC2, MTTP, and APOB in rat groups Compared to the normal group, immunoreactivity of CDX2, MUC2, MTTP, and APOB increased in the model group rats. In the group treated with naringenin and morin, immunoreactivity of CDX2, MUC2, MTTP, and APOB decreased compared to the model group ($n = 6$, randomly select 3 fields of view for each sample). (Scale bar = 50 μm).



from *C. reticulata* Blanco. This result indicated that QLF may regulate IM mainly by naringenin.

Molecular docking analysis was then performed to simulate interactions of naringenin with MTTP and APOB (Figure 4). It was indicated that APOB can form traditional hydrogen bond interaction forces with naringenin through their own Phe-2, Gln-42, Cys-44, Ser-45, and Glu-89, and MTTP can form traditional hydrogen bond interaction forces with naringenin through its own Asn-550 and Gln-663. In addition, there were hydrophobic interactions and C-H hydrogen bonds between the two, and under the action of these interaction forces, the binding energy of APOB and Naringenin was -7.7 kcal/mol, and the binding energy of MTTP and Naringenin was -7.6 kcal/mol, both of which performed well. These results suggested that naringenin may regulate IM progression by targeting MTTP and APOB.

3.5 | Naringenin suppressed MTTP and APOB expression in GES-1 cell line

In order to establish the IM model, we used DCA to treat the immortalized gastric epithelial cell line GES-1. Figure 5a, b showed that DCA-treated GES-1 cells expressed more CDX2 and MUC2 than control cells,

indicating the success of the IM model. There was also an increase in the mRNA expression of MTTP and APOB (Figure 5c, d). MTTP/APOB expression can be inhibited by naringenin alone. A significant reduction in MTTP/APOB expression was also observed in the naringenin-treated group compared to control group. As a result, naringenin may have a therapeutic effect on IM through its inhibition of the MTTP/APOB pathway.

To validate the role of the MTTP/APOB pathway in the process of IM, we designed MTTP siRNA and examined the impact of MTTP knockout on CDX2 and MUC2. In DCA-treated GES-1 cells, the mRNA levels of CDX2 and MUC2 increased significantly, while MTTP knockout markedly inhibited this process (Figure 6a, b). Furthermore, MTTP knockout significantly reduced the levels of APOB mRNA in DCA-treated GES-1 cells (Figure 6c, d). These findings suggested that the MTTP gene plays a critical role in the progression of IM and that the inhibitory effect of naringenin on IM through the MTTP/APOB pathway may be feasible.

3.6 | Naringenin regulated MNNG-induced IM model through the MTTP/APOB pathway in rat

The model was successfully established in the 24th week, and the immunohistochemistry showed a significant increase in CDX2 and MUC2 levels in the model group compared to the control group. After

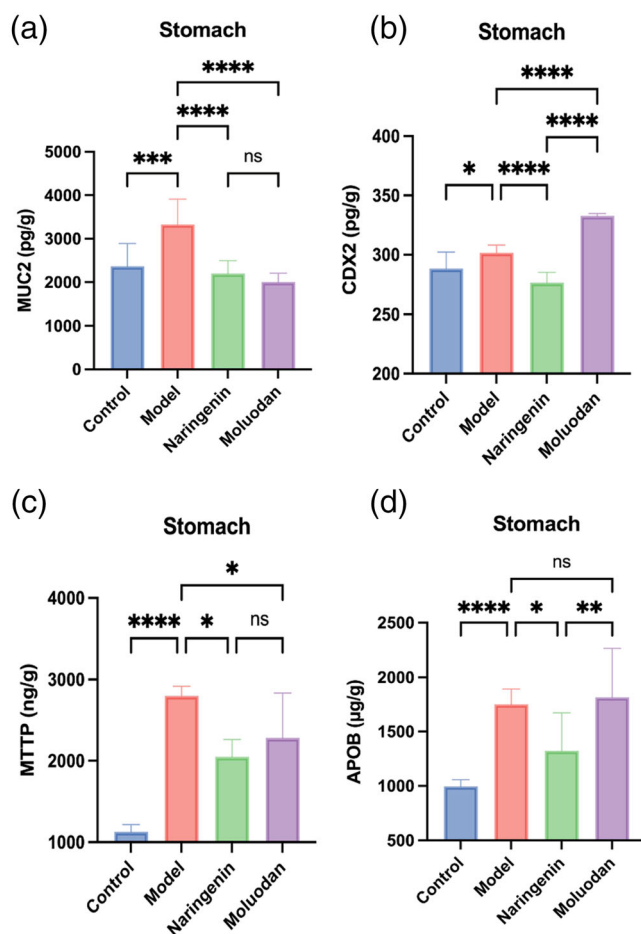


FIGURE 8 Protein levels of CDX2, MUC2, MTTP, and APOB in gastric tissues of rats from each group. Compared to the normal group, the protein levels of CDX2, MUC2, MTTP, and APOB were increased in the model group, while they were decreased in the groups treated with naringenin and moluodan ($n = 6$, data were presented as the mean $\pm$ SEM, all experiments had been repeated at least three times). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: $P \geq 0.05$.

8 weeks of treatment with naringenin and moluodan by gavage, the levels of CDX2 and MUC2 in the stomach were significantly reduced, and the levels of MTTP and APOB decreased accordingly (Figure 7). The results in gastric tissue and serum were similar. The protein levels of CDX2, MUC2, MTTP, and APOB in the model group were significantly higher than those in the control group. Naringenin and moluodan could restore the levels of these proteins. In the gastric tissue, the protein levels of CDX2 and APOB in the naringenin group were significantly lower than those in the moluodan group, indicating that the effect of naringenin was better than moluodan (Figures 8 and 9).

Due to the crucial role of APOB in the generation and function of low-density lipoprotein (LDL), its deficiency or malfunction may lead to disturbances in lipid metabolism. Therefore, we further examined in rat models the serum concentrations of four lipoproteins: triglycerides (TGs), total cholesterol (TC), LDL, and high-density lipoprotein (HDL). As seen in Supplementary Figure 5, naringenin could decrease the

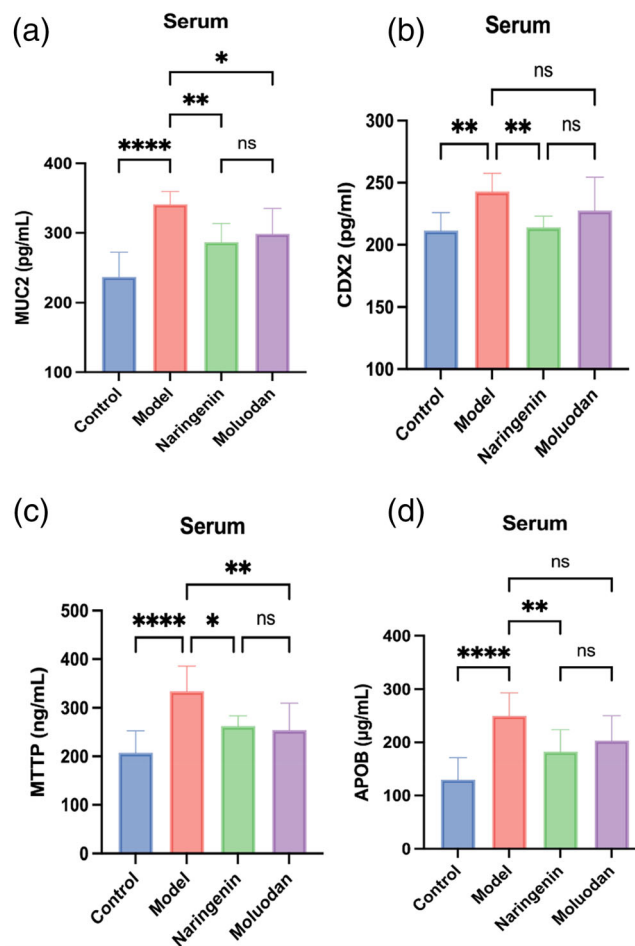


FIGURE 9 Serum levels of CDX2, MUC2, MTTP, and APOB in rats from each group. Compared to the normal group, the serum levels of CDX2, MUC2, MTTP, and APOB were increased in the model group, while they were decreased in the groups treated with naringenin and moluodan ($n = 6$, data were presented as the mean $\pm$ SEM, all experiments had been repeated at least three times). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: $P \geq 0.05$.

concentration of TG, TC, LDL-C, and HDL-C compared to the model group. Additionally, APOB plays an important role in physiological and pathological processes such as immune regulation, inflammatory response, and apoptosis. Therefore, we further detected the expression levels of four cytokines, IL-6, IL-1 β , Casp-3, and TNF- α , in the serum of rats in each group. It indicated that naringenin significantly reduced these cytokines. Taken together, these results suggested that naringenin may regulate IM through the MTTP/APOB pathway.

4 | DISCUSSION

The treatment of IM with traditional Chinese medicine has become increasingly popular in recent years (Lou et al., 2024; Tang et al., 2016; Xie et al., 2024; Zhang et al., 2018; Zhou et al., 2022). One herbal constituent that is commonly used in this practice is naringenin, which belongs to the flavanone group of polyphenols and is

found in citrus fruits such as lemons, blood oranges, and tangerines, as well as grapes (Motallebi et al., 2022). This phytochemical has shown antioxidant, anti-inflammatory, and anti-cancer activity, and both *vitro* and *in vivo* studies demonstrate naringenin's therapeutic potential for gastric conditions such as ulcers and GC (Bao et al., 2016; Faramarzi et al., 2022; Park et al., 2017). In this study, we identified several hub genes associated with IM progression by combining the RNA-seq results of clinical samples with publicly available datasets. Additionally, we demonstrated that naringenin influenced IM development through the regulation of *MTTP*/*APOB*.

Notably, we meticulously considered the drug concentrations in both animal and cell experiments. In the animal studies, the concentration of naringenin was 50 mg/kg/day, and the concentration of Molodan was 10 mL/kg/day. These doses were referenced from the animal-human dose conversion formula for potential translation in rats (Nair & Jacob, 2016). In cell experiments, we tested the effects of naringenin on cell viability at concentrations ranging from 0 to 20 mmol, ultimately selecting 10 mmol. Regarding the potential toxicity of naringenin, it is generally considered safe, with most studies reporting low toxicity. However, its poor water solubility poses a challenge for its therapeutic application, limiting its bioavailability (Nouri et al., 2019; Salehi et al., 2019).

As part of our study, we set relatively strict criteria for DEG screening. The common DEGs were compared with the WGCNA results, and 30 genes were chosen. Based on GO and KEGG pathway enrichment analysis, these genes were primarily involved in lipid transport. Additionally, fat digestion and absorption, cholesterol metabolism, and *PPAR* signaling pathways were also enriched. All functions appeared to be closely related to digestion and absorption, suggesting that changes in the stomach's digestive function and eating habits are closely linked to the progression of IM to cancer. In recent studies, the incidence of IM has been linked to alcohol consumption and high-salt or spicy diets, and it is believed that abnormal lipid metabolism was an independent risk factor for gastric cancer (Jayavelu & Bar, 2014). As the *PPAR γ* protein appears to have anti-tumor properties and is expressed in normal mucosa adjacent to tumor sites (Sato et al., 2000), a strategy to prevent IM from evolving into cancer via targeting this pathway may be feasible.

Based on molecular docking and qRT-PCR experiments, *MTTP* and *APOB* were identified as potential targets of naringenin. Chylomicrons, very-low-density lipoprotein (VLDL), and low-density lipoprotein (LDL) are produced with *APOB*, which is the primary protein in these particles. Thus, *APOB* plays a crucial role in lipid metabolism. *MTTP* facilitates lipoprotein synthesis by transferring lipids to *APOB* proteins and regulates the assembly and secretion of *APOB*-containing lipoprotein. In most cases, this is due to its ability to transport lipid components (TGs, cholesterol esters, and phospholipids) to the endoplasmic reticulum lumen where lipoproteins are assembled. Patients with gastric cancer with high levels of *APOB* expression have shorter overall survival, and *APOB* levels are negatively correlated with distant metastases (Zhang et al., 2016). By reducing *MTTP* and *APOB* mRNA levels, naringenin improves lipid

metabolic imbalance; specifically, *MTTP* mRNA levels are significantly reduced by naringenin, thereby inhibiting *APOB*-*MTTP* binding, which results in *APOB* degradation. It is thought that naringenin inhibits *APOB*-mediated HCV secretion primarily by blocking *APOB* secretion and VLDL and LDL assembly (Hernández-Aquino & Muriel, 2018). When *MTTP* inhibitors are used *in vivo*, total cholesterol, LDL-C, TG, VLDL cholesterol, and *APOB* levels can be effectively reduced. Furthermore, these inhibitors may have anti-atherosclerotic and insulin-sensitizing properties. Lastly, the *PPAR* signaling pathway also appears to play an important role in naringenin's regulation of *APOB*, as *PPAR α* inhibition has been shown to reverse this effect (Goldwasser et al., 2011).

Despite this study's strengths, there are still some limitations that must be noted. First, we used a small number of clinical samples. However, our results were enhanced via the use of publicly-available datasets. There was also a lack of experimental studies that provided detailed information. Thus, further study is needed to more thoroughly elucidate and validate the mechanism by which naringenin regulates IM progression through *MTTP* and *APOB*.

5 | CONCLUSIONS

This study reports for the first time that naringenin inhibits the *MTTP*/*APOB* axis to inhibit IM progression. The results of our study support the development of naringenin as an anti-IM agent based on theoretical and experimental evidence and may contribute to the discovery of novel IM therapeutic targets.

AUTHOR CONTRIBUTIONS

Xiangming Huang: Software; validation; visualization; writing – original draft. **Mengqiu Zhang:** Software; validation; visualization; writing – original draft. **Lina Gu:** Data curation; investigation; validation. **Ziyan Zhou:** Software; validation; visualization; writing – original draft. **Shengtong Shi:** Formal analysis. **Xinyu Fan:** Formal analysis. **Wei Tong:** Formal analysis. **Dazhi Liu:** Formal analysis. **Jihu Fang:** Formal analysis. **Xinen Huang:** Resources. **Zhijun Fang:** Conceptualization; funding acquisition; methodology; project administration; supervision; writing – review and editing. **Min Lu:** Conceptualization; funding acquisition; methodology; project administration; supervision; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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